

Mitigating lead toxicity towards safer commercialization of perovskite solar cells

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Metal halide perovskite solar cells (PSCs) have emerged as transformative photovoltaic technologies due to their high efficiency and low cost. However, the presence of lead raises environmental and health concerns, posing a barrier to commercialization. This Review quantifies the toxicity risks associated with lead in PSCs and underscores the need for effective mitigation strategies. We discuss the long-term effectiveness of lead isolation and immobilization approaches by introducing quantitative metrics, specifically the sequestration efficiency and lifetime. We further evaluate the practical feasibility of these strategies in terms of device efficiency, scalability and cost. In addition, we explore sustainable lead sources and advocate for the leveraging of lead–acid battery recycling as a transitional supply, ultimately aiming to establish a closed-loop lead recycling system for PSCs. By integrating lead-sequestration strategies, establishing recycling infrastructure for end-of-life perovskite photovoltaics and advancing producer responsibility policies, this Review provides a roadmap towards the safer and more sustainable commercialization of lead-based perovskite photovoltaics.

Metal halide perovskites have drawn intense interest in photovoltaics. Over the past decade, perovskite solar cells (PSCs) have reached a power conversion efficiency (PCE) of 27.0% (ref. 1). Owing to their tunable bandgap, numerous perovskite-based tandem architectures have emerged. Notably, perovskite/silicon tandem configurations have achieved certified efficiencies of 34.6% for small-area devices, surpassing the best silicon single-junction performance (~27.8%)². For industry-scale third-party-certified modules, Trina Solar have achieved a PCE of 27.2% (ref. 3) for perovskite/silicon tandem modules at 1.52 m², and GCL Optoelectronic Materials have reached a PCE of 19.04% (ref. 4) for single-junction perovskite modules with an area of 2.0 m², highlighting the substantial progress in large-area fabrication. Stability has also improved under continuous illumination⁵, high-temperatures⁶ and

reverse bias⁷. However, despite these advances in large-area efficiency and stability, the industrialization of perovskite photovoltaics still requires clear and enforceable standards, particularly regarding the use of lead (Pb). Before deployment at scale, a complete governance framework should be established to track lead across the life cycle and to define, monitor and mitigate leakage pathways.

Lead is highly toxic yet remains crucial for achieving high efficiency and stability in PSCs. Whereas substituting lead with tin (Sn) or germanium (Ge) could reduce toxicity, the PCE and stability are severely limited by Sn²⁺ and Ge²⁺ oxidation^{8,9}. Previous estimates have indicated that a 1 GW installation of 20% efficient perovskite modules with a 500 nm absorber would contain about 3.5 tons of lead¹⁰. Consequently, the potential release of lead from perovskite solar modules

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during operation or at end of life has raised widespread concerns due to its harmful effects on humans, animals, plants and microorganisms^{11–14}. Despite substantial progress on mitigating Pb²⁺ leakage and lead recycling, the real-world deployability and policy alignment for perovskite solar modules remain insufficiently established, blurring the commercialization roadmap. In parallel, recycling should move towards a perovskite-to-perovskite closed-loop system with a clearly defined transitional pathway.

In this Review we quantify the toxic effects of lead at different stages of the cradle-to-cradle design of perovskite photovoltaics through life-cycle assessment (LCA), to emphasize the importance of suppressing Pb²⁺ leakage from perovskite photovoltaics. To mitigate the hazards associated with lead toxicity from perovskite photovoltaics, we discuss three approaches that ease policy restrictions and facilitate the commercialization of lead-based PSCs: the use of low-lead/lead-free perovskites, suppressing Pb²⁺ leakage and lead recycling. We focus on evaluating various innovative lead-sequestration strategies by introducing the sequestration efficiency (SQE) lifetime as a key quantitative indicator, and assess the practical implications of these strategies in terms of device efficiency, scalability and cost. Regarding lead recycling, we propose leveraging the recycling of lead–acid batteries (LABs) as a transitional lead source towards a closed-loop system for PSCs. Finally, we suggest integrating lead-related LCA metrics into extended producer responsibility (EPR) policies to bridge the gap between science assessment and policy design for sustainable PSCs.

Quantifying the risk of lead in perovskite photovoltaics

The environmental impact of lead in PSCs is closely linked to its content, chemical composition and release mechanisms. The systematic quantification of Pb²⁺ exposure pathways throughout the PSC life cycle, along with the development of experimental assessment protocols, is essential for formulating effective strategies to mitigate lead toxicity in the perovskite photovoltaics industry.

Life-cycle assessment of lead in PSCs

Lead is a bioaccumulative toxicant that affects multiple organ systems. In 2003, a blood concentration for lead of 5 µg dl^{−1} was set as the clinical reference for children¹¹. Subsequent evidence of cognitive and behavioural impairment at lower exposures led to the withdrawal of this threshold in 2010. Adults may also suffer neurological deficits from chronic or high-dose acute exposure. Current studies have confirmed that lead exhibits genotoxicity and damages hematopoietic and reproductive systems¹⁵ (Fig. 1a). The toxicity of lead is strongly influenced by the solubility of its compounds. Under environmental stress, perovskites degrade to more soluble lead species that increase the bioavailability and mobility of Pb²⁺. In plants, perovskite-derived Pb²⁺ is readily absorbed and accumulates in the roots and leaves. Low-level exposure elevates the tissue lead burden, whereas higher doses cause blackening and decay (Fig. 1b)¹³. In microbial systems, lead-based perovskites inhibit the growth of *Escherichia coli*, whereas tin-based perovskites exhibit no adverse effect (Fig. 1c)¹⁴.

The LCA evaluates environmental impacts and sustainability within defined system boundaries. Early LCA studies on perovskite photovoltaics adopted cradle-to-gate boundaries that emphasized energy consumption and environmental hotspots during raw material extraction and manufacturing¹⁶. Subsequent work expanded the system boundary to cradle-to-grave, incorporating the use and disposal phases to better approximate full life-cycle sustainability. Recent analyses introduced recycling and remanufacturing and moved the systems boundary towards a cradle-to-cradle approach that follows the raw material extraction, manufacturing, use, recycling and remanufacturing cycle, with evaluations expanded to tandems and module-level systems (Fig. 1d)^{17–20}. As layer-by-layer perovskite photovoltaics enable easy separation and recycling, evaluating their

sustainability within a cradle-to-cradle system boundary offers a more realistic outlook on future industrial practices. Under harmonized functional unit and cradle-to-gate conditions, PSCs demonstrate lower environmental impacts than c-Si (crystalline silicon), CdTe and CIGS (copper indium gallium selenide), owing to their low-temperature processing²¹. When recycling and remanufacturing are incorporated under cradle-to-cradle analyses, delamination-friendly perovskite photovoltaics can shorten the energy payback time (EPBT) by about 72.6% and reduce the greenhouse gas (GHG) emission factor by about 71.2% (ref. 22). In all-perovskite tandem devices, recycling can reduce the GHG emission to 4.64 g CO₂e MJ^{−1}, lower than that of silicon photovoltaics (4.73 g CO₂e MJ^{−1}), and the energy return on investment can reach 15.0, higher than the benchmark value of 14.8 for silicon photovoltaics¹⁸. It should be noted, however, that these benefits depend strongly on the recycling methods used. Processes that involve highly toxic solvents may offset environmental gains²³, whereas the use of green solvents or solvent recirculation can enhance overall sustainability^{24–26}. For instance, lead recycling pathways that use *N,N*-dimethylformamide and ethyl acetate show higher global warming potential and toxicity than water-based methods. However, by reusing the solvent over ten cycles, the global warming potential of these solvent-intensive pathways can be reduced by approximately 89.55% (ref. 27).

Electrodes and transparent conductive oxides have been identified as major environmental hotspots in PSCs²⁸. Whereas lead-free perovskites seem less hazardous by elemental toxicity, LCA shows they may not lower the overall environmental impact²⁸. Toxicity alone is therefore inadequate for assessing sustainability. If Pb²⁺ leakage is effectively controlled, lead-based PSCs can outperform other photovoltaics in GHG and energy metrics. Nevertheless, the toxicity and persistence of lead remain critical barriers to sustainable commercialization because of regulation and public acceptance. A complete assessment of lead toxicity should use cradle-to-cradle boundaries and explicitly include failure scenarios. Using the USES LCA method (where USES is a multi-media fate, exposure and effects model that is based on the Uniform System for the Evaluation of Substances) with electricity generation as the functional unit, analyses indicate that Pb²⁺ leakage across the life cycle is the primary driver of both the ecotoxicity and the human toxicity potential (Fig. 1e,f)²⁹. Under the US Resource Conservation and Recovery Act (RCRA) framework, the toxicity characteristic leaching procedure (TCLP) sets a regulatory limit for total lead at 5 mg l^{−1}. However, standard leaching tests for PSCs report 14.2 ± 0.2 mg l^{−1} from glass and 713 ± 5 mg l^{−1} from polyethylene terephthalate substrates, which classifies the materials as hazardous waste and raises disposal costs by about 75-fold³⁰. Lead is both bioaccumulative and persistent in the environment. Even minimal release can lead to its accumulation through soil, water and food chains, with risks heightened by insufficient regulatory oversight. Given these risks, the stringent containment of lead throughout the life cycle of the device is imperative.

Pathways of environmental contamination from damaged modules

Trace lead can be released during normal operation, even when the encapsulation appears intact. Long-term weathering, combined with ultraviolet (UV) radiation and thermal ageing, accelerates perovskite degradation and creates diffusion pathways³¹. Under damp-heat conditions, moisture can infiltrate along module edges and layer interfaces, even in the absence of visible mechanical damage³². Mechanical stress induces microcracks that serve as rapid channels for humidity and ion penetration³³, whereas surface abrasion compromises the protective barrier, exposing the functional stack to water and runoff³⁴. More severe damage, for example, impacts from flying debris, hail or exposure to extreme weather events such as storms and fires, can rupture the encapsulation, leading to pronounced Pb²⁺ release, particularly during rainfall. The leakage process generally occurs in two stages:

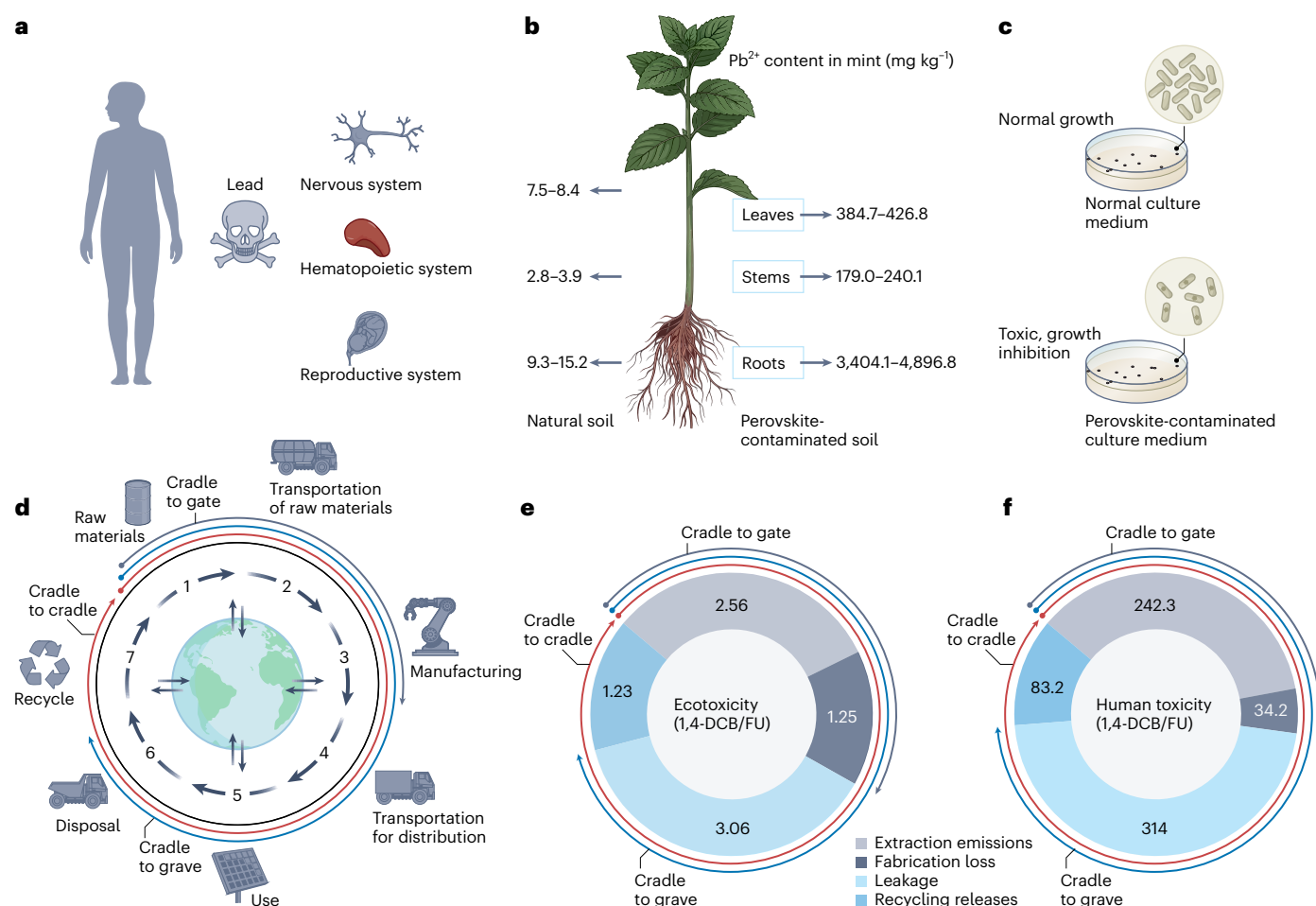


Fig. 1 | Lead toxicity of PSCs. a, Toxic effects of lead on the human body. **b**, Total lead in mint plants cultivated on natural soil versus perovskite-contaminated soil. **c**, Schematic showing *E. coli* grown in control versus perovskite-contaminated media. **d**, System boundary definitions for the LCA of PSCs, including cradle-to-gate (raw material extraction to factory gate), cradle-to-cradle (production, use

and end of life) and cradle-to-cradle (closed-loop recycling and remanufacturing) boundaries. **e, f**, Life-cycle ecotoxicity (**e**) and human toxicity (**f**) attributable to lead in PSCs. The system boundary is cradle to cradle, and the functional unit is 1 kWh of electricity generation. 1,4-DCB/FU, 1,4-dichlorobenzene per functional unit.

initial release from the damaged PSCs and subsequent leaching into groundwater or soil.

In the initial release stage, water rapidly disrupts the perovskite lattice, forming soluble lead halides that dissolve and drive sustained Pb^{2+} release through a diffusion-dominated process. The diffusion process can be modelled using a semiempirical approximation derived from the Noyes–Whitney equation, which has been applied to study metal leakage from damaged PSCs:

$$\frac{dm}{dt} = A \times \frac{D}{d} \times (C_s - C_b), \quad (1)$$

where m represents the mass of dissolved lead, t denotes the diffusion time, A is the effective exposure area (a critical factor that influences the Pb^{2+} leakage rate), D is the diffusion coefficient of Pb^{2+} , d is the thickness of the diffusion layer, C_s is the saturated mass concentration at the solution surface and C_b is the mass concentration in the bulk solution. The Pb^{2+} leakage rate can be estimated by calculating A (ref. 35). This rate is influenced by factors such as the temperature, pH value and ion type. For example, Pb^{2+} leakage rates in water at 85 °C are found to be 7.9 times greater than those at 45 °C. Nitric acid-type acid rain (pH = 4.0) causes leakage rates that are seven times and 20 times higher than sulfuric acid-type acid rain (pH = 4.0) and deionized water (pH = 7), respectively³⁶.

The second stage, that is, environmental diffusion into rivers, groundwater or soil, depends on the physicochemical properties of the environmental media. In water, the environmental diffusion of Pb^{2+} can be described using the Noyes–Whitney equation. In soil, however, Pb^{2+} diffusion and ecotoxicological effects are largely governed by its chemical speciation and the physicochemical characteristics of the soil matrix. Lead compounds show a clear toxicity hierarchy³⁷, and Pb^{2+} from halide perovskites exhibits approximately threefold higher bioavailability compared with existing underground lead pollution sources, thus highlighting the challenges in establishing safe exposure thresholds¹³. Moreover, chemical speciation in soil substantially influences its diffusion behaviour. In carbonate- or phosphate-rich soils, Pb^{2+} tends to form low-solubility compounds ($PbCO_3$, with a solubility product constant (K_{sp}) of 1.46×10^{-13} , and $Pb_5(PO_4)_3OH$, with a K_{sp} value of $10^{-76.8}$) that restrict their mobility. In addition, organic cations from perovskites can alter the soil pH, further affecting Pb^{2+} diffusion and bioavailability¹³.

Methods for the assessment of lead leakage

Establishing reliable evaluation methods for Pb^{2+} leakage is critical for quantifying environmental impacts and health risks during the commercialization of perovskite modules. Outdoor or rooftop-installed modules face extreme weather (for example, heavy rain, hail), which can cause physical damage and accelerate Pb^{2+} leakage. Building on the

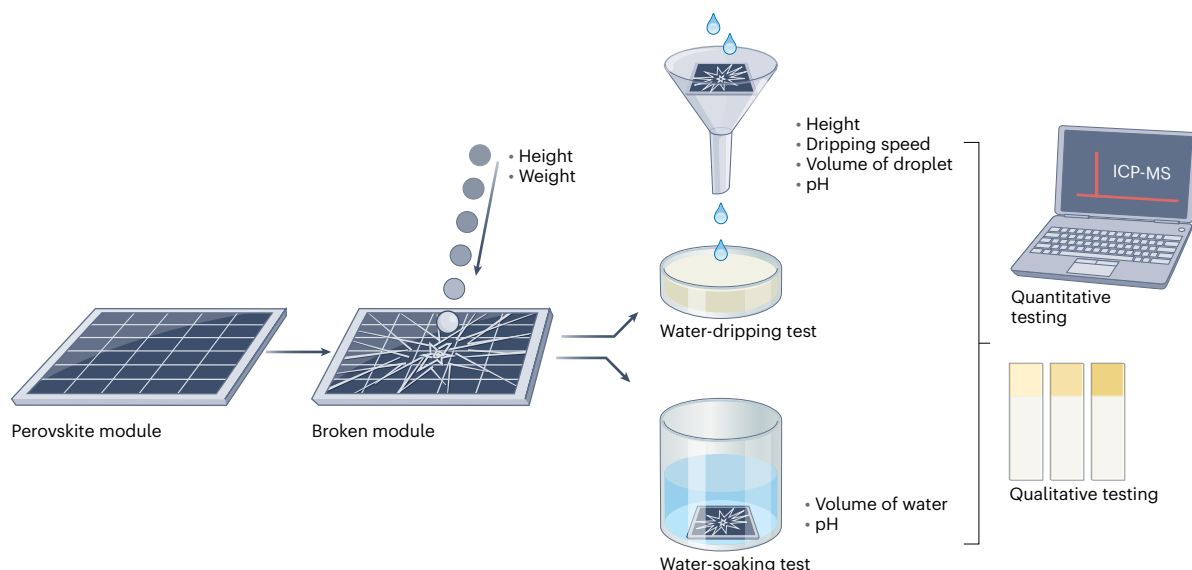


Fig. 2 | Assessment of lead leakage for perovskite solar modules. Hail impact is simulated to induce damage, after which the damaged modules are rinsed or soaked in water, and Pb^{2+} leakage is quantified using ICP-MS or test strips.

conventional standard for rigid photovoltaic modules (FM 44787), a specialized Pb^{2+} leakage assessment protocol was developed for perovskite modules. The method simulates hail impacts by dropping standardized metal spheres on to light-receiving surfaces from calibrated heights³⁵. To simulate rainwater rinsing, a syringe pump then rinses the modules with deionized or acidic water at a controlled rate. The Pb^{2+} leakage rate is then analysed via inductively coupled plasma mass spectrometry (ICP-MS), which provides a high sensitivity of trace Pb^{2+} concentrations across a broad dynamic range. For rapid field assessments, test strips offer an approximate quantification of aqueous Pb^{2+} levels (Fig. 2). By incorporating regional annual precipitation data, the potential extent of lead contamination can be roughly estimated. However, variations in rainfall patterns and rainwater pH across different regions may affect the leakage severity, leading to differing environmental risks. Furthermore, because Pb^{2+} release from perovskites is time-dependent, assessments must include prolonged rainwater exposure to ensure accurate evaluations.

Reducing lead toxicity in PSCs

Following quantification of the toxic risks of lead, we turn to mitigation and lowering the regulatory burdens. Using lead-free or lead-reduced perovskites addresses the hazard at its source, provided that PCE and stability meet the requirements. If lead-based perovskites are used, risk must be reduced by suppressing leakage while ensuring that lead is contained in a closed-loop system through traceable collection and recycling to prevent environmental release.

Lead-free and lead-reduced perovskites

Reducing or eliminating the lead content directly alleviates public concern and eases regulatory pressure. Partial substitution with tin in lead–tin alloys has delivered certified efficiencies exceeding 24.13% while retaining over 90% of the initial performance after 795 h under maximum power point tracking⁸. Pure-tin-based perovskites benefit from high charge-carrier mobility and narrow bandgaps, and have achieved a PCE of 17.9% (ref. 38). However, they still face performance degradation due to the oxidation of Sn^{2+} to Sn^{4+} (ref. 39). Whereas reductive additives⁴⁰, two-dimensional (2D) capping layers⁴¹ and surface dipoles⁴² can partially mitigate these issues, stability and reproducibility remain limiting. Germanium and tin–germanium alloy perovskites have efficiencies of around 7% or less due to oxidation instability⁹.

Toxicology and performance should be evaluated within the same framework. Lead-free compositions reduce intrinsic toxicity but do not

mean more sustainability. When electricity generation is used as the functional unit in LCA, device efficiency and operational lifetime dominate toxicity metrics, energy use and EPBT. Devices with single-digit efficiencies or lifetimes of between one and five years generally fail to offer advantages in GHG emissions or overall energy consumption⁴³. Some lead-free alternatives may also require greater material inputs, higher-energy deposition processes or highly toxic solvents, resulting in higher toxicity than lead-based perovskites²⁸. Lead-based perovskites remain a necessary baseline for comparison. Single-junction lead-based devices now exceed 27% efficiency, yet their environmental risks and social acceptability still depend on the effective suppression of Pb^{2+} leakage. Lead–tin alloys can reduce the lead inventory with modest performance penalties and support high-efficiency tandem devices, which helps to ease regulatory and societal concerns. However, LCAs of state-of-the-art lead–tin alloyed perovskite photovoltaics have not yet been systematically performed within a unified framework that uses device area or electricity generation as the functional unit. Such assessments would help to establish environmental and economic benchmarks relative to lead-based perovskites.

Mitigation of Pb^{2+} leakage in lead-based perovskites

Lead halide perovskites are highly sensitive to water. When damaged devices are exposed to rainwater, the highly polarized H_2O molecules form strong hydrogen bonds with organic cations and halides. This disrupts the bonding between the A-site cations and the $[\text{PbI}_6]^{4-}$ octahedron, triggering irreversible perovskite lattice collapse and rapid decomposition into PbI_2 (Fig. 3a). Owing to the high solubility constant of PbI_2 ($K_{\text{sp}} = 8.49 \times 10^{-9}$ at 25 °C), it readily dissolves into mobile Pb^{2+} in water, increasing the risk of environmental contamination (Fig. 3a). To alleviate this risk, strategies for mitigating Pb^{2+} leakage can be established either by blocking water ingress or immobilizing dissolved Pb^{2+} . As these strategies target different degradation phases, their mechanisms and design principles vary substantially.

Isolation strategies for mitigating Pb^{2+} leakage. Isolation strategies aim to enhance perovskite hydrophobicity or improve film quality to resist moisture-induced degradation (Fig. 3b,c). Researchers commonly incorporate low-polarity, highly hydrophobic functional groups, such as alkyl chains ($-\text{C}_n\text{H}_{2n+1}$)^{44–47}, thiols ($-\text{SH}$)^{48,49}, trifluoromethyl groups ($-\text{CF}_3$)^{50,51}, aromatic rings ($-\text{C}_6\text{H}_5$, phenyl group)^{52–54} and siloxanes ($-(\text{SiR}_2\text{O})_n-$)⁴⁷, to form a moisture-protective barrier at the

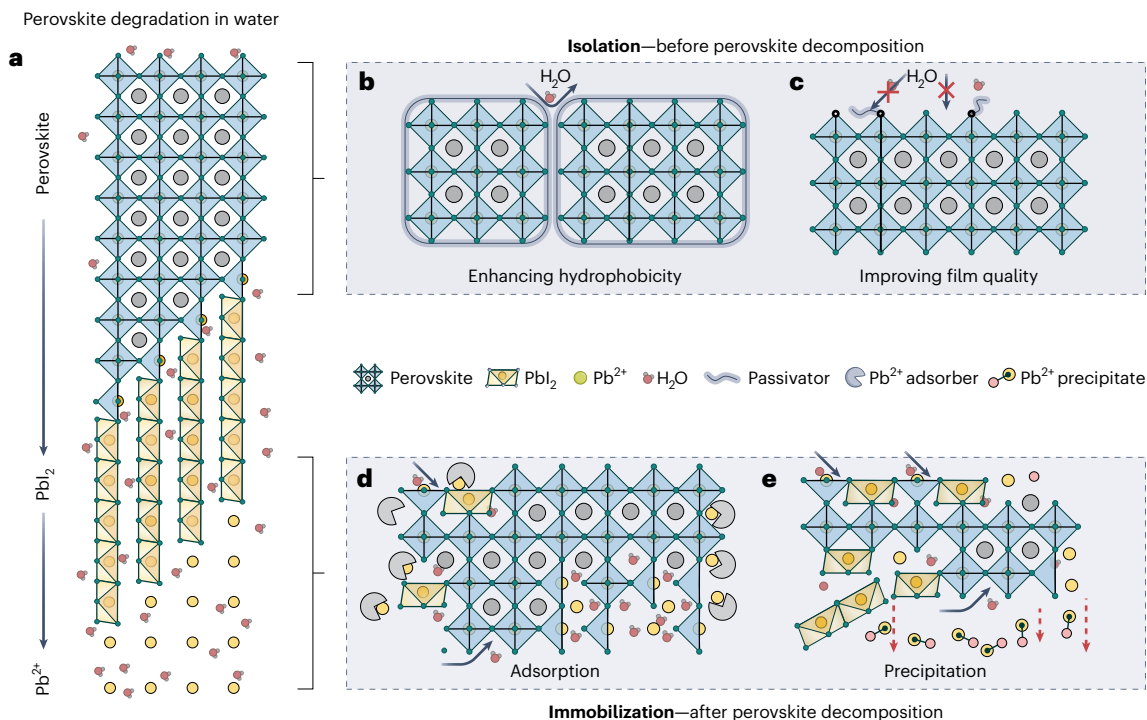


Fig. 3 | Strategies and mechanisms for mitigating Pb^{2+} leakage in PSCs. a, Dissolution process of perovskite in water. **b–e**, Fundamental mechanisms of isolation (**b,c**) and immobilization (**d,e**) strategies.

perovskite grain boundaries or interfaces. These functional groups weaken interfacial interactions with water molecules, reducing water adsorption and enhancing material stability. Hydrophobic molecules are integrated through three key approaches: (1) dimensional reduction to induce 2D perovskites⁴⁵, (2) additive incorporation at grain boundaries to enable grain encapsulation⁵⁵ and (3) the formation of dense hydrophobic overlayers at interfaces⁵³. The effectiveness of these strategies depends not only on the intrinsic hydrophobicity of the functional groups but also on the uniformity and thickness of the protective layer. Particular attention has been paid to enhancing film uniformity and coverage. For instance, cross-linking reactions enable hydrophobic materials to form a dense network structure, creating a more uniform and continuous protective layer on the perovskite surface, which effectively suppresses water infiltration^{51–54,56}.

Water-induced perovskite decomposition typically initiates at defect sites, where adsorbed water molecules progressively penetrate the lattice and cause structural collapse. Therefore, improving the quality of perovskite crystallization to reduce defects and enhance inherent water resistance has become a key focus. Many hydrophobic additives feature functional groups that simultaneously regulate crystal growth and passivate defects, enabling dual benefits of suppressing Pb^{2+} leakage and enhancing device performance^{50,51,57}. Despite these advantages, internal isolation strategies face inherent limitations. Whereas effective in the short term, their protective capacity diminishes under prolonged water exposure or immersion. A key challenge is in the dosing of hydrophobic materials, as insufficient loading fails to provide adequate protection whereas excessive loading degrades device performance. Moreover, physical damage to perovskite modules can fracture the film, expose untreated surfaces and dramatically increase the risk of Pb^{2+} leakage.

Immobilization strategies for mitigating Pb^{2+} leakage. Immobilization strategies capture dissolved Pb^{2+} during perovskite decomposition through adsorption or precipitation (Fig. 3d,e). Adsorption-based immobilization relies on functional groups that

form strong coordination bonds with Pb^{2+} , including sulfonic acid ($-\text{SO}_3\text{H}$)^{58–60}, amino ($-\text{NH}_2$)^{61,62}, phosphate ($-\text{PO}_4$)^{63,64} and carboxyl ($\text{C}=\text{O}$) groups^{14,65}. The efficacy of these functional groups depends on the coordination strength, adsorption site density and material morphology. NMR spectroscopy has been used to determine the binding constant between functional groups and Pb^{2+} , offering insight into the coordination strength⁶⁶. In addition, density functional theory calculations have been used to evaluate the binding energies of different functional groups with Pb^{2+} . Their results show that $-\text{CONH}$ binds with Pb^{2+} at 3.04 eV and $-\text{NH}_2$ at 2.87 eV. The combination of the two functional groups ($-\text{CONH}-\text{R}-\text{NH}_2$) achieves the highest binding energy of 3.3 eV, enhancing the Pb^{2+} adsorption capability⁶². Besides, material morphology plays a crucial role in adsorption efficiency. Gel-like cation-exchange resin (CER) shows a superior adsorption capacity compared with porous CER, due to its higher surface area and site accessibility^{58,67}. Precipitation immobilizes Pb^{2+} by forming low- K_{sp} water-insoluble lead compounds. Common additives include sulfonic acid groups^{68,69}, carbonyl groups⁷⁰, porphyrin rings⁷¹ and phosphoric acid⁶⁴. Thermodynamically, precipitation is driven by decreased Gibbs free energy, favouring the formation of stable and insoluble compounds. A lower K_{sp} value indicates a more spontaneous precipitation reaction⁷². However, the stability of the precipitate is strongly influenced by the stability of the binding sites and the environment. Lowering the pH or exposure to competing ions enhances the redissolution kinetics and gradually releases Pb^{2+} , leading to secondary pollution⁷³. Besides, repeated wet–dry cycling, freeze–thaw, and mechanical abrasion generate fissures and fresh surfaces, shifting the dissolution–reprecipitation balance and allowing fine lead-containing solids to enter environmental media in colloidal or particulate forms⁷⁴. Therefore, mitigation measures should focus on improving the stability of the precipitate to reduce secondary pollution. To ensure long-term Pb^{2+} immobilization, researchers have focused on developing pH-independent precipitation materials. Recently, a large-area textured phosphate-buffered functionalized polymer film (PFPF) with self-healing properties was designed to immobilize Pb^{2+} through

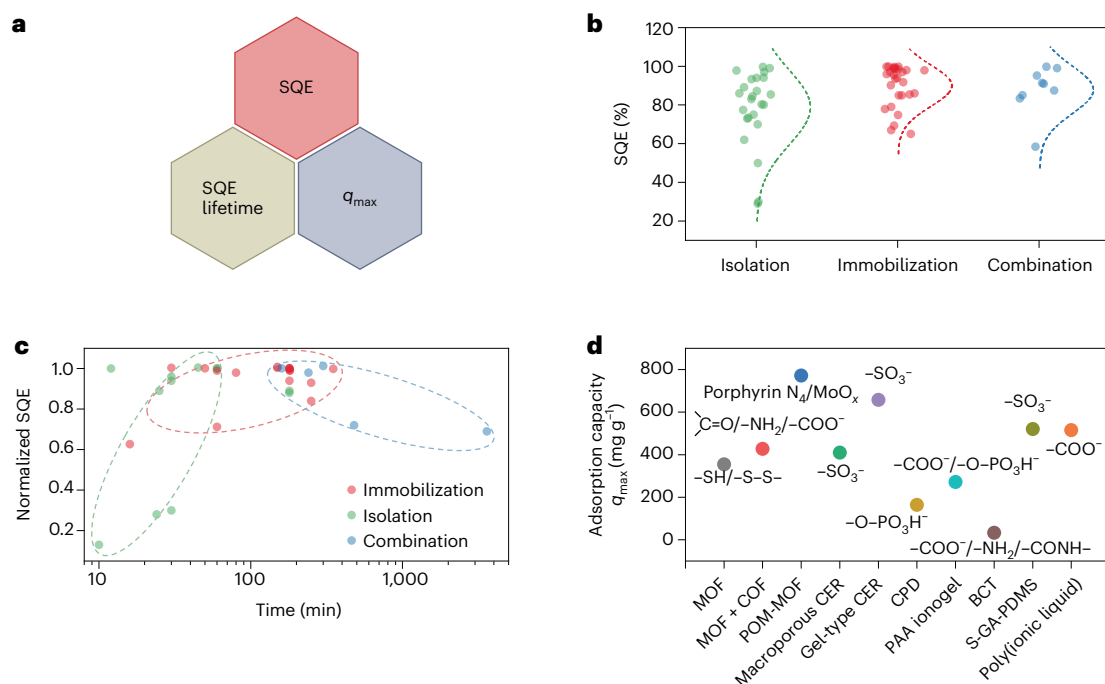


Fig. 4 | SQE and stability. **a**, Key performance metrics for Pb²⁺ leakage mitigation. **b,c**, SQE (**b**) and SQE lifetime data (**c**) for Pb²⁺ leakage suppression using isolation^{44,45,51,53,57,84,85,111,113,115,116,118}, immobilization^{58–60,62,63,69–71,75,81,88,103–105} and their combination^{76–80}. **d**, Maximum Pb²⁺ adsorption capacities of different materials and functional group

ps^{58,60–62,65,67,76,81,102,103,117,120}. POM, polyoxometalate; CPD, crosslinked composite obtained via chemical principles-based design; PAA, poly-acrylic acid; BCT, bioinspired cage trap; S-GA-PDMS, mixed sulfonated graphene aerogel and polydimethylsiloxane.

phosphate precipitation. The PFPF stabilized the local pH, enabling an ultrafast Pb²⁺-sequestration rate across a wide range of environmental pH conditions⁶⁴.

Immobilization materials can be incorporated into PSCs via internal embedding or exterior encapsulation. For embedding, Pb²⁺-adsorbing materials can be added to the perovskite layer or charge-transport layers. However, to avoid performance degradation, doping concentrations are typically kept below 1% (refs. 59,68,69), and the additional layers are limited to a thickness below 10 nm (refs. 66,75). These constraints limit the number of available adsorption sites for Pb²⁺ immobilization. By contrast, external encapsulation with Pb²⁺-adsorbing materials enables thicker protective layers (often in the micrometre range), thereby providing a much larger number of adsorption sites for Pb²⁺ sequestration.

Performance of Pb²⁺ leakage mitigation. The SQE, SQE lifetime and maximum adsorption capacity ($q_{\max}$) are the three key metrics for quantifying Pb²⁺ leakage mitigation in PSCs (Fig. 4a). The SQE was proposed as a standardized metric⁶³, and it is defined as:

$$\text{SQE} = 1 - \frac{\text{Pb}_{\text{leakage from devices with Pb absorbers}}}{\text{Pb}_{\text{leakage from devices without any Pb absorbers}}} \times 100\% \quad (2)$$

We compare the SQE values of different mitigation strategies, including isolation, immobilization and their combined strategies. The results show that immobilization and combination exhibit higher and more consistent SQE values than isolation alone (Fig. 4b). This is because most isolation strategies primarily focus on enhancing the hydrophobicity of the perovskite or transport layers. To avoid efficiency losses, the amount of hydrophobic material used is typically low, limiting its effectiveness.

SQE durability is a key metric because environmental stresses are unpredictable. Natural disasters or mechanical impacts may

compromise module integrity and trigger Pb²⁺-leakage events that may continue for hours or days before professional remediation. Such scenarios demand not only rapid initial suppression but also prolonged containment efficacy to mitigate ecological and human health hazards. Analysis of normalized SQE trends from the literature shows distinct temporal behaviours among the Pb²⁺ mitigation strategies. The SQE of isolation exhibits a sharp decline within 30 min (refs. 44,45,51,53), whereas immobilization remains relatively stable for tens to hundreds of minutes^{58,60,62,63}. The combination of both strategies demonstrates the best durability, often maintaining a high SQE for several hundred minutes (Fig. 4c)^{76–80}. This trend is logical because isolation strategies do not capture Pb²⁺, allowing progressive leaching over time and consequent SQE decline. By contrast, immobilization continuously removes Pb²⁺ by capturing it, ensuring a stable SQE. When isolation and immobilization are combined, Pb²⁺ release is delayed while adsorption occurs simultaneously, leading to the most stable SQE. This highlights the importance of integrating both strategies to mitigate Pb²⁺ leakage.

The $q_{\max}$ metric can be used to directly evaluate the potential of a material to suppress Pb²⁺ leakage. It is defined as the maximum mass of Pb²⁺ adsorbed per unit mass of material and can be obtained via Langmuir model fitting. As shown in Fig. 4d, a comparative analysis of $q_{\max}$ values reported in the literature reveals substantial variations that are influenced by functional group characteristics (type and number) and structural parameters (porosity and specific surface area). For example, via the in situ homogeneous growth of the β -ketoenamine-linked covalent organic framework TpPa-1-COF on the metal-oxide framework MOF-808, MOF@COF core-shell nanoparticles were obtained that achieved a $q_{\max}$ of 427.9 mg g⁻¹, which is attributable to the high density of Pb²⁺ adsorption sites and to the large specific surface area and pore size⁸¹. In addition, among materials with similar functional groups, a gel-like CER exhibited a $q_{\max}$ of 657.89 mg g⁻¹, which higher than the 410 mg g⁻¹ of the porous CER^{58,67}. Functional group chemistry further dictates the performance variations among gel-type materials.

For instance, carboxyl and phosphate groups in a polyacrylic acid ionogel ($q_{\max} = 271.43 \text{ mg g}^{-1}$) show a lower adsorption capacity than the sulfonate-containing gel-like CER^{55,67}.

The parameter $q_{\max}$ is independent of the placement (encapsulation or embedding), sample state (intact device, film or fragments) and testing conditions. This independence makes $q_{\max}$ a direct and reliable metric for evaluating a material's intrinsic Pb^{2+} adsorption capability, offering critical insights for selecting optimal adsorbents. As a key equilibrium adsorption parameter, $q_{\max}$ also facilitates cross-material comparisons, guiding rational materials design. By optimizing the surface chemistry, structural features and functional group distribution, materials can be engineered to achieve a superior Pb^{2+} adsorption performance, ultimately enhancing their effectiveness in Pb^{2+} leakage mitigation.

Practical implications of the lead-sequestration strategy. Although many materials exhibit a high SQE, their practical implications require holistic evaluation. Key considerations include their impact on the photovoltaic efficiency, scalability and production costs, and whether their effectiveness in suppressing Pb^{2+} leakage meets the relevant policy requirements to ensure technical and economic viability.

The efficiency depends on where the Pb^{2+} -sequestering materials are placed in the stack. To remove placement effects, we analysed embedding and encapsulation separately and computed the change in efficiency ΔPCE via $\text{PCE}_{\text{after}} - \text{PCE}_{\text{before}}$. With embedding, across p–i–n and n–i–p architectures (where p, n and i represent p-doped, n-doped and intrinsic semiconductors, respectively) and across isolation and immobilization chemistries, ΔPCE is typically positive with similar medians, consistent with crystallization control, interfacial defect passivation and improved charge transfer when additives are embedded^{14,52,82}. However, whereas higher loading is more effective in suppressing Pb^{2+} leakage, it also decreases the PCE. Therefore, the additive concentration should be optimized to achieve effective Pb^{2+} leakage suppression without compromising the device performance. For external encapsulation, the efficiency is governed by optical transmittance. Front-side encapsulation can reduce the PCE through reflection or parasitic absorption, whereas high transmittance or antireflection designs tend to be optically neutral or even slightly beneficial (Fig. 5a).

Laboratory spin coating enables precise additive placement and often yields high PCE and SQE values. However, under high-throughput manufacturing, achieving uniform nanoscale coverage across large modules becomes more challenging, and the SQE performance may decline. Encapsulation is inherently more scalable because it is decoupled from perovskite crystallization and enables flexible thickness control, although its efficiency gains are typically smaller than those achieved through embedding. Recent demonstrations have shown that blade coating, inkjet printing and physical vapour deposition can effectively deposit Pb^{2+} -sorbing layers after perovskite crystallization or directly on to finished devices, maintaining film integrity and offering good compatibility for large-scale production^{61,83}. The data in Fig. 5b illustrate this trade-off: embedding clusters at higher ΔPCE with a moderate SQE^{51,54,56,84,85}, whereas encapsulation delivers a high SQE at moderate ΔPCE ^{60,65,70,86,87}. For scalability, embedding should emphasize additive uniformity, interactions with the precursor and tolerance to the crystallization window. Encapsulation should emphasize scalable deposition of the sorbent layer, optical management on the front side, resistance to weathering and mechanical stress, and chemical compatibility with established encapsulants (Fig. 5c).

The economic feasibility of suppressing Pb^{2+} leakage is central to commercialization. As illustrated in Fig. 5d, polyolefin elastomer (POE) and ethylene vinyl acetate (EVA), widely used in commercial photovoltaic modules, are priced at US\$0.95 per m^2 and US\$0.86 per m^2 , respectively. However, these materials lack any intrinsic Pb^{2+} -adsorption capacity and do not sufficiently mitigate Pb^{2+} leakage in perovskite photovoltaics. Earlier Pb^{2+} -adsorption materials, such as self-healing epoxy

resin (US\$3.08 per m^2), *N,N,N',N'*-ethylenediaminetetrakis(methylene phosphonic acid) (EDTMP; US\$11.8 per m^2) and *P,P'*-di(2-ethylhexyl) methanediphosphonic acid (DMDP; US\$441.2 per m^2)⁶³, are more expensive than conventional encapsulants. The higher costs arise from their complex synthesis, expensive raw materials or limited manufacturing scale. In recent years, several lower-cost Pb^{2+} -adsorption materials have been developed, including crosslinked fluoropropyl methylsiloxane-dimethylsiloxane multiblock polymer (CFDP; US\$1.26 per m^2)⁸⁶, sulfonated SiO_2 mixed with polyvinyl alcohol (SSP; US\$1.73 per m^2)⁸⁸ and PFPFs (US\$2.018 per m^2)⁶⁴. These materials combine strong Pb^{2+} adsorption with an SQE above 99% and offer pricing that is competitive for large-scale applications. Even so, the cost of most Pb^{2+} -adsorption encapsulation materials remains higher than that of POE and EVA, mainly because of their limited production scale and market demand. In addition to developing low-cost and scalable Pb^{2+} sorbents, a practical alternative is to functionalize commercial encapsulants with Pb^{2+} -binding moieties such as phosphate, carboxyl or amino groups. This often requires only an additional step in material synthesis and can preserve the cost advantage of existing materials while remaining compatible with current photovoltaic encapsulation infrastructure, which can accelerate practical deployment.

Lead recycling

This section compares recycling pathways for perovskite devices and used LABs, assessing the device performance, environmental burdens and technological maturity for each route. It also considers recycling logistics and policy support to achieve closed-loop recycling for perovskite photovoltaics.

Recycling pathways and process assessment. For PSCs, the lead supply comes mainly from three pathways: (1) primary lead mining, (2) recycling from perovskites and (3) recycling from LABs, as shown in Fig. 6. In natural deposits, lead occurs mainly as galena, anglesite and cerussite⁸⁹, with an average crustal abundance of 16 ppm (ref. 90). According to data from the International Lead and Zinc Study Group, over 80% of global refined lead consumption in 2022 was used for the production of LABs, whereas only about 5% was used as lead compounds that can serve as precursors for perovskite synthesis (Fig. 6)⁸⁹. Conventional primary production relies on roasting and reduction at temperatures above 1,400 °C, releasing carbon dioxide, sulfur dioxide and lead-containing vapours or particulates, which impose severe environmental burdens⁹¹. With tighter environmental regulations and the advance of circular economy strategies, many countries are reducing their reliance on primary lead. In the USA, the recycling of LABs supplied about 70% of lead consumption in 2024⁹². Thus, establishing efficient and scalable lead recycling routes that are capable of producing high-purity secondary lead has become a key strategy to alleviate environmental pressure and mitigate reliance on primary resources.

Lead recycling from perovskites can be categorized into dissolution–precipitation (D–P)^{25,93} and adsorption–separation (A–S)^{62,78}, as illustrated in pathway (2) of Fig. 6. In the D–P method, perovskite layers are dissolved using polar organic solvents or acidic aqueous media, then an antisolvent or iodide is added to precipitate lead as PbI_2 or perovskite powders^{25,94–96}. The A–S route first releases Pb^{2+} using polar solvents and subsequently captures it with adsorbents such as zeolites, polymers, ion-exchange resins or bioinspired materials^{14,62,97}. Desorption and iodide-assisted conversion enable further separation and purification^{10,14,62}. However, lead recycling from perovskites remains at the proof-of-concept stage and lacks industrial-scale processes. By contrast, LAB recycling is mature and scalable, and is supported by extensive infrastructure and policy frameworks. Conventional LAB recycling involves carbonate treatment and high-temperature reduction to extract metallic lead, followed by acid leaching and iodination to obtain PbI_2 (pathway (3) in Fig. 6).

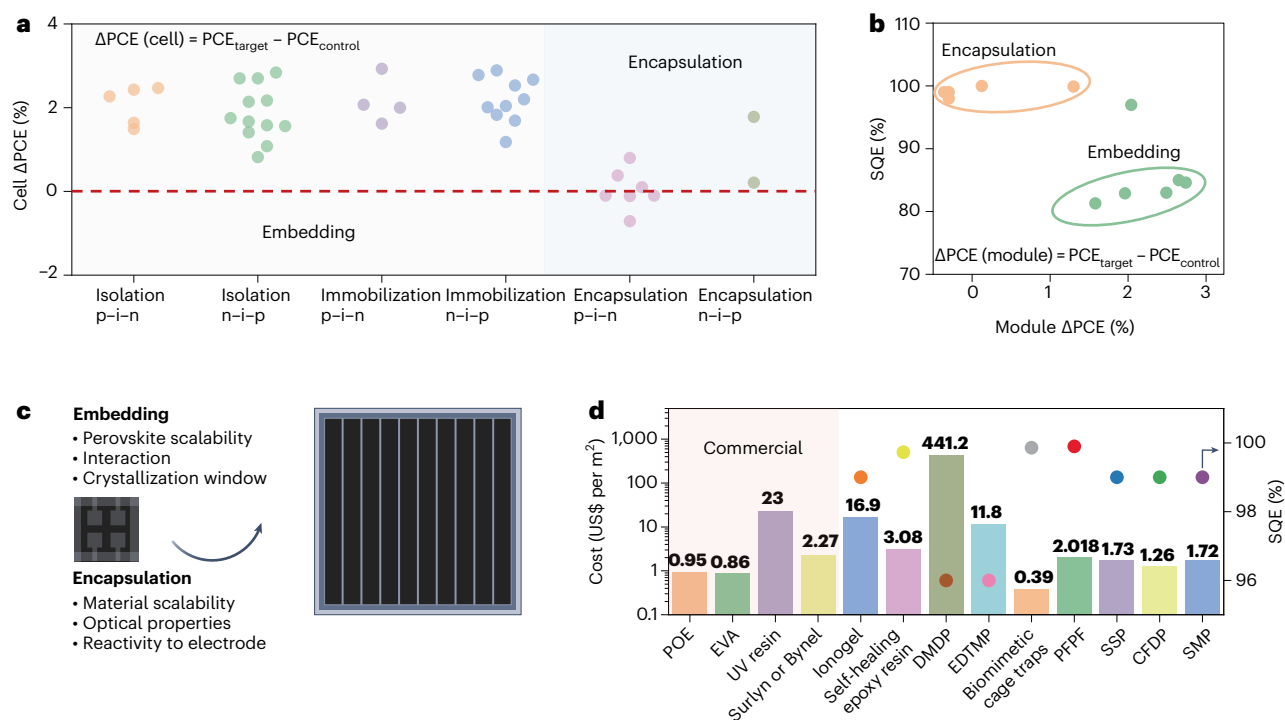


Fig. 5 | Practical implications of the lead-sequestration strategy.

a, Impact on the solar cell performance after implementing isolation via embedding^{45,46,48,49,51–54,56,57,84,109,112,114,116,118,126–128} or immobilization via embedding^{12,14,59,61,62,66,68,70,71,81,102,106,107,129} or encapsulation^{38,63–65,83,86,103,104}, respectively. We restricted the analysis to embedded devices with an initial PCE above 20% to enable a meaningful comparison. **b**, Trade-off between the change

in module efficiency and SQE for embedding^{21,54,56,84,85} and encapsulation^{60,65,70,86,87}. **c**, Key considerations for scaling when integrating Pb²⁺-adsorbing materials via embedding or encapsulation. **d**, Cost comparison of scalable Pb²⁺-adsorbing encapsulants versus commercial encapsulants, together with SQE for the former^{35,62–65,86,88,103}. All cost values are normalized to 2025 US dollars for a meaningful comparison. SMP, sulfosuccinic-acid-modified polyvinyl alcohol.

We compared PSCs fabricated with lead recycled from perovskites or LABs with those using purchased lead, and their PCE values are comparable, with ΔPCE typically within $\pm 1\%$ (Fig. 6 inset plot^{14,93,98–100}). These results show that recycled lead can meet the purity requirements for high-efficiency device fabrication. Moreover, most lead recycling yields exceed 95%, with some achieving over 98%. However, most perovskite recycling routes rely on toxic and volatile solvents that are costly to treat, imposing environmental and economic burdens at scale. Therefore, low-toxicity, water-compatible or biodegradable solvent systems are urgently needed for safer and more sustainable lead recycling.

Recycling institutional considerations. Lead recycling from perovskite photovoltaics is constrained not only by recycling processes but also by deployment patterns and supply chain structure. Unlike centralized photovoltaic power plants, distributed perovskite photovoltaics are deployed across rooftops, vehicles and consumer electronics. These systems are highly dispersed, ownership is fragmented and many units fall outside formal collection channels, allowing them to enter municipal waste streams without proper handling. In addition, the lead content in photovoltaic devices is relatively low, and the encapsulation structures vary across applications, making centralized treatment difficult and economically unattractive. This situation contrasts sharply with LABs, which have simple structures and well-established reverse logistics systems. Therefore, beyond developing efficient and environmentally sound lead recycling technologies, system-level and policy measures are essential. Key priorities include implementing EPR schemes, promoting standardized and disassemblable device designs, and integrating perovskite photovoltaics into existing photovoltaic and electronic waste collection networks. Such measures can strengthen governance frameworks and help to establish a resilient, circular and environmentally sustainable perovskite photovoltaic industry.

Sustainable lead sources. Advances in recycling technologies and policy frameworks can enable practical lead recovery from perovskite photovoltaics and support the development of sustainable lead supply chains. However, the current volume of end-of-life perovskite modules remains too small to meet near-term industrial demand. A phased roadmap is therefore required to balance immediate feasibility with long-term sustainability. To establish a sustainable lead supply chain and promote green manufacturing, we propose that in the early stage of industrialization, perovskite photovoltaics could utilize lead recycled from LABs as an interim supply source. This approach would reduce reliance on primary mining while taking advantage of the mature LAB recycling infrastructure. In the longer term, a dedicated closed-loop lead recycling system within the perovskite industry should be established to achieve sustainable lead resource management and to promote the green, low-carbon development of perovskite photovoltaic technology.

Relevant policy

In addition to stability and scalability, the industrialization of lead-containing perovskite photovoltaics must comply with policy and regulatory requirements. In the European Union, EPR and end-of-life governance anchor the policy landscape, and photovoltaic modules fall under the Waste Electrical and Electronic Equipment directive¹⁰¹. In the USA, beyond the federal RCRA framework, several states are moving ahead with photovoltaic-specific EPR. China has issued a new mandatory China RoHS (Restriction of Hazardous Substances) standard, GB 26572-2025, which strictly limits market access for electronic products containing lead. Public acceptance is equally pivotal. The 'lead' label triggers concern even when the lead content is low. Addressing doubts about the hazards of lead before market deployment is essential. Robust and reliable Pb²⁺-sequestration technology is therefore critical. Under

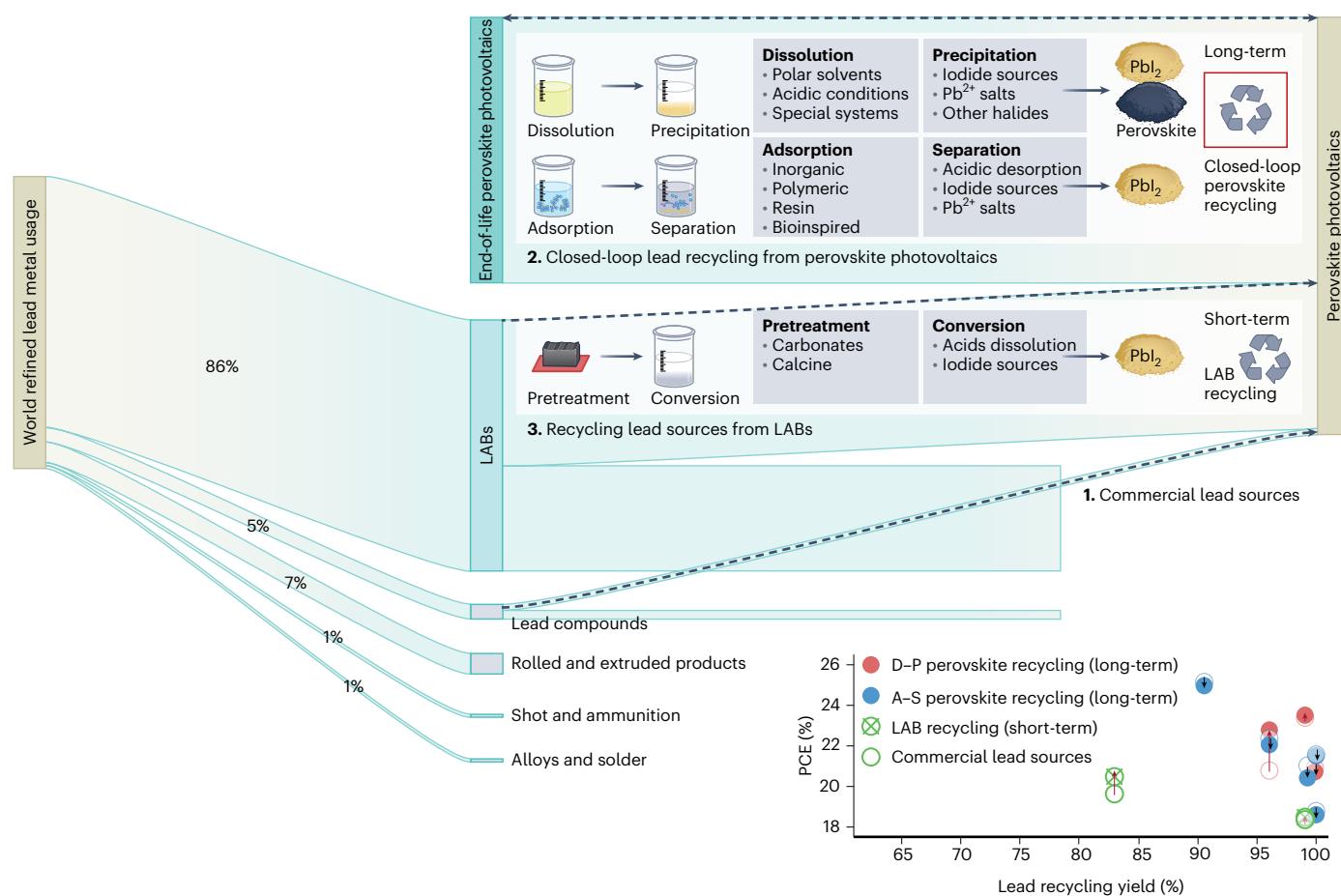


Fig. 6 | Lead recycling. Pathways of refined lead inflows into perovskite photovoltaics, lead recycling strategies and their effects on device performance and recovery yield. Data sources: LAB recycling^{100,130}, D-P perovskite recycling^{25,93,98}, A-S perovskite recycling^{10,62,78,97,131}.

the TCLP of the RCRA framework, samples are subjected to an acidic environment placed in an acidic medium and agitated for 18 ± 2 h followed by the measurement of total lead. This standard implies that a reliable sequestration should show minimal dependence on pH and sustained stability. At present, there are no dedicated standards for evaluating Pb²⁺ release from perovskite photovoltaics. Most assessments use water-drip tests or immersion tests, and neither long-term stability nor pH dependence has become a fixed requirement. Establishing unified scenario tests that incorporate extreme weather and post-damage conditions—including long-term Pb²⁺ release and secondary pollution—and set leakage thresholds linked to market access would enhance both regulatory credibility and public confidence.

LCA provides a quantitative framework for linking environmental performance with policy implementation. By translating environmental burdens into standardized indicators (for example, toxicity potential per unit power), LCA can inform differentiated EPR fee structures, recycling targets and leakage performance standards. For instance, modules that show lower life-cycle toxicity or higher recyclability could qualify for reduced EPR fees or extended recycling credits, establishing an incentive mechanism that rewards pollution reduction. In addition, LCA can map experimental Pb²⁺ leakage results on to life-cycle toxicity indicators, transforming physical concentration thresholds into measurable environmental impact factors for compliance monitoring. This approach enables regulatory thresholds to reflect not only chemical concentration limits but also real environmental impacts across a product's lifetime. Embedding LCA metrics into EPR and recycling frameworks can therefore create a feedback loop between scientific assessment and policy design, paving the way for a data-driven,

traceable and verifiable closed-loop management system for lead in perovskite photovoltaics.

Conclusion and outlook

Lead has enabled perovskite photovoltaics to reach record efficiencies, yet it remains a major obstacle to their commercialization because of the risk of Pb²⁺ leakage. Implementing innovative strategies to suppress Pb²⁺ leakage and developing sustainable lead sources are critical for reducing environmental and health risks. A roadmap for safer commercialization emphasizes material design, combined mitigation strategies and policy frameworks that support sustainable deployment.

We summarize the mechanisms, performance impacts and implementation potential of Pb²⁺ leakage mitigation strategies (Table 1). Currently, hybrid approaches that combine isolation and immobilization show great potential for improving the SQE and stability. However, achieving uniform nanoscale coverage over large embedded areas is a practical challenge. Encapsulation materials that immobilize Pb²⁺ should retain strong binding across a wide pH window and in ligand-rich environments, provide a high site density at low loading and withstand UV light, heat and humidity to prevent secondary pollution. For cost considerations, beyond developing low-cost, scalable Pb²⁺ sorbents, a practical approach is to integrate Pb²⁺-binding functionalization into commercial encapsulants. This preserves the cost advantage of the encapsulant, remains compatible with existing photovoltaic encapsulation infrastructure and accelerates practical deployment.

Beyond preventing Pb²⁺ leakage, recycling lead modules during their operational lifetime or at end of life is equally important. EPR with traceable take-back is essential for distributed photovoltaics.

Table 1 | Decision matrix for Pb²⁺ leakage mitigation strategies

Category	Immobilization		Isolation
	Embedded	Encapsulation	Embedded
Mechanism	Complexation or precipitation to capture soluble Pb ²⁺		Improve perovskite stability and reduce moisture ingress
Functional groups	Phosphate, sulfonic, carboxyl, thiol/disulfide and other Lewis-base donors coordinating Pb ²⁺		Hydrophobic: fluorinated, siloxane, 2D perovskites Film-quality enhancement: Lewis acid–base passivation and crystallization regulation
Period	During or after leakage: adsorb or chelate Pb ²⁺ once perovskite decomposes		Pre-leakage: suppress moisture ingress before perovskite decomposes
SQE	High		Moderate
Placement	Perovskite bulk/surfaces, interfaces and electrode side	Rear-side, front-side and edge seals	Perovskite bulk/surfaces, interfaces, electrode side
SQE lifetime	Moderate	High	Low to moderate
ΔPCE	+1 to +3%	−0.5 to +1.5%	+1 to +3%
Scalability	Challenge in achieving uniform nanoscale coverage at module scale	Good	Challenge in achieving uniform nanoscale coverage at module scale

There is evidence that devices fabricated with recycled lead from perovskite photovoltaics show no obvious performance loss, even though this technology is not yet as mature as LAB recycling. Because the installed base of perovskite photovoltaics remains small, early sustainable supply can rely on recycled lead from LABs. As retired perovskite modules accumulate, a perovskite-to-perovskite closed-loop system can be established. This transition window provides time for perovskite recycling technologies to mature and become more environmentally benign. Together, effective sequestration and a sustainable supply framework provide a clear and safe roadmap for lead-safe perovskite photovoltaics.

Data availability

All data are available in the main text. Source data are provided with this paper.

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Author contributions

D.L., Y.J. and Y.Q. jointly contributed to the conceptualization, literature analysis, figure preparation and writing of the manuscript. Y.H. contributed to the discussion of the recycling institutional considerations and assisted in revising the manuscript. Q.C. and Z.Y. provided support for the manuscript revision and the response to reviewers. F.Y. contributed to the discussion of the LCA content. All authors were involved in discussions regarding data analysis and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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